# Modelling Student Academic Success: A Knowledge Graph Approach Integrating Socioeconomic, Behavioural, and Educational Factors

Tiago Rodrigues[1[64375]], João Ascenso[1[56939]], Rui Costa[1[29280]], and Manuel Gonçalves[1[58555]]

[1] Faculdade de Ciências da Universidade de Lisboa, Lisboa 1749-016, Portugal
{fc64375, fc56939, fc29280, fc58555}@alunos.fc.ul.pt

**Abstract.** Academic performance is influenced by a wide range of socioeconomic, behavioural, and educational factors. In this project, we present the design and implementation of a Student Academic Success Knowledge Graph that semantically models the interactions among these variables. Using data from the Kaggle Student Alcohol Consumption dataset, we represent entities such as students, schools, courses, parents, and social behaviours as classes within a custom ontology, and define their relationships through structured object and datatype properties. The knowledge graph captures complex factors including family background, study habits, health status, and extracurricular involvement. Leveraging this graph-based representation, we aim to predict students' final grades by integrating domain knowledge with machine learning methods. Our approach demonstrates how knowledge graphs can enhance predictive modelling and support data-driven educational research by enabling intelligent querying, inference, and discovery of relevant academic patterns.

**Keywords:** Knowledge Graph · Ontology Modelling · Grade Prediction

## 1 Introduction

Academic success is a multifaceted outcome of a knotted interaction of socioeconomic status, behavioural tendencies, intellectual aptitudes, and learning milieus. It is not only a reflection of students' intellectual capacity but also of a wide range of external and internal conditions, including family environment, peer relationship, study habits, health status, and institutional support. It is a fundamental challenge for researchers, policymakers, and also teachers attempting to improve educational outcomes and close achievement gaps to know how these complex factors interact and influence academic success.

Modelling academic achievement conventionally has, on a large scale, relied on classical statistical techniques and machine learning algorithms on tabular data that tend to deal with each attribute in isolation or with very little relational

context. While such methods have yielded valuable information, they are ill-suited to capture the complex, relational nature of educational processes. For example, a student's success is not only a function of personal attributes but also of the quality of interactions with family members, school environment, and peer groups. These sorts of relationships introduce dependencies and contextual nuances difficult to represent in two-dimensional data structures.

Knowledge graphs (KGs) provide a promising paradigm to surmount these limitations by representing information as a graph of entities and relationships capturing the rich semantic interdependencies of educational data. By casting students, schools, parents, and study habits as nodes interconnected by meaningful edges, KGs enable the holistic modelling of the educational ecosystem. The graph-based formalism enables semantic querying, logical reasoning, and the extraction of complex patterns that may remain opaque under traditional data structures.

Latest advances in graph embedding techniques provide a way to embed knowledge graphs into continuous vector spaces while preserving their structural and semantic properties. These embeddings allow for the application of machine learning algorithms that can take advantage of latent features from graph topology and attributes. Integration of knowledge graph embeddings with predictive analytics holds much promise to improve the accuracy and interpretability of student performance prediction.

In this paper, we construct a *Student Academic Success Knowledge Graph* from the publicly available Kaggle Student Alcohol Consumption dataset with complete information on students' academic performance, family, behaviour, and health indicators. We define an ontology that delineates key concepts such as *Student*, *School*, *Parent*, and *StudyHabit*, and represents the semantic relations among these entities. We apply embedding methods to extract meaningful latent features from this graph and use them to train machine learning models for predicting students' final grades.

Our approach addresses key issues in educational data mining by bridging the divide between symbolic semantic representations and predictive modelling driven by data. It offers a unified framework that enhances not just predictive power but also delivers interpretable insights on the multifactorial determinants of academic success. This synthesis of machine learning and semantic technologies assists in the development of intelligent, explainable educational analytics software that can support evidence-based decision-making.

The contributions of this paper are as follows:

1. We design a comprehensive (however simple) knowledge graph ontology for the multifaceted factors affecting student academic performance.
2. We leverage advanced graph embedding techniques to capture semantic relationships and discover latent features for prediction.
3. We demonstrate the power of knowledge graph embeddings combined with machine learning models for predicting student final grades with competitive results.

4. We derive understandable conclusions from the semantic model that specify prominent factors influencing academic performance.
5. We validate our approach on a widespread real-world dataset, demonstrating its feasibility and promise for educational analytics.

This paper is structured as follows: Section 2 covers related work on educational data mining, knowledge graphs, and graph embeddings. Section 3 describes the dataset, ontology design, and knowledge graph creation. Section 4 describes the embedding models and predictive modeling framework. Section 5 presents experimental results and analysis. Section 6 concludes the paper and gives future research directions.

## 2 Related Work

Academic performance of students has been the subject of extensive interdisciplinary work, drawing from areas such as educational psychology, data mining, machine learning, and semantic web technologies. In this section, we cover prior work on three general topics of interest to our project: statistical and machine learning techniques, graph-based student education models, and ontology-based modeling of student achievement.

### 2.1 Statistical and Machine Learning Methods

Initial predictions of student performance were primarily statistical modeling-based, which identified single, independent factors influencing academic achievement, such as socioeconomic status, parental involvement, and homework study habits [6]. These initial models, while groundbreaking, did not often possess the ability to identify complex, multivariate interactions between factors.

With the advent of educational data mining, more sophisticated machine learning methods have been applied to model academic achievement. Techniques such as decision trees, logistic regression, support vector machines (SVM), and neural networks have been applied using datasets to predict levels of student dropout, grade attainment, and levels of engagement [4] [1]. Although these models are more predictive in performance, they process data in a two-dimensional flat table format, discarding the rich contextual and relational information that may influence the performance of students. Furthermore, machine learning models have the threat of explainability issues, which render them less clear and applicable in real-world educational practice.

### 2.2 Graph-Based Data Modeling of Education

In recent years, graph-based data modeling—using knowledge graphs specifically—has gained growing traction in the field of education. Knowledge graphs enable entities (e.g., students, courses, institutions) to be described as nodes and

their relations (e.g., enrollment, performance, social contact) as edges, providing a flexible, semantically rich setting for reasoning and querying.

Such initiatives as EduKG [7] have demonstrated the effectiveness of knowledge graphs to connect and fuse heterogeneous educational data, including learner data, learning material, and pedagogical activities. Graph models enable adaptive recommendations and personalized learning based on dependencies and relations that are difficult to specify working with conventional relational or flat data models.

In addition, the use of graph embeddings, techniques that learn vector representations of elements in graphs, has enabled the application of machine learning to graph-structured data, enhancing student performance prediction and early warning systems [2].

### 2.3 Ontology-Driven Models for Education

Ontology-based modeling in education provides an interoperable and formal way of representing domain knowledge. Ontologies define concepts, relationships, and constraints in a structured manner, allowing shared understanding, semantic reasoning, and integration across systems.

Some of the most relevant examples are the Open University Linked Data project [3], which described linked semantic data with aims to facilitate educational resource discovery, curriculum mapping, and metadata annotation. Similarly, the S3O (Student Semantic Success Ontology) [5] defined an ontology that was specifically designed to capture the dimensions of student engagement, behavioral patterns, and academic trajectories.

Such ontology-based systems possess several unique strengths in the aspects of context-aware modeling, reusability of knowledge, and semantic query support based on standards such as OWL and SPARQL. They enable the development of extensible and transparent smart educational systems.

## 3 Methods

This section describes the methodology used to construct a semantic knowledge graph based on learning data. The process involves four major phases: data preprocessing, development of the ontology, building knowledge graphs, and predictive modelling.

### 3.1 Dataset and Preprocessing

We employed the `student-mat.csv` data from the Kaggle Student Alcohol Consumption dataset, which provides information on 395 students at a secondary school who were studying mathematics. The dataset comprises varied attributes that range from demographic details to academic history, family circumstances, lifestyle habits, and three grades attained in the academic year: `G1`, `G2`, and `G3`. The last term grade, `G3`, has been employed as the predictand variable.

Limited preprocessing was applied to preserve the raw semantics and dataset structure. The following was done:

- Parsing the CSV file in Python (Pandas) to process student records.
- Raw attribute values to people and data literals in the ontology mapped directly without label encoding or label normalization.
- Building individuals for category properties (e.g., school, address, guardian) and relating them with students through object properties.
- Stating numerical properties (e.g., age, absences, G1, G2, G3) as data property values by applying RDF literals.
- Omitting null or missing values to maintain RDF triple consistency.

This approach retains the original data set and resulting RDF representation close in synchrony, facilitating a semantically rich and interpretable knowledge graph without invoking radical transformation or abstraction.

### 3.2 Ontology Design

We constructed an ontology specific to the domain of formally representing entities and relations. The first ontology was programmatically built with Python and the `rdflib` library to enable immediate integration with the dataset and entity instantiation automation. The ontology was subsequently imported into Protégé to polish the structure, check consistency, and increase semantic definition.

Core classes include *Student*, *School*, *Parent*, *HealthStatus*, and *StudyHabit*. Relationships among entities were modelled using object properties such as *attends*, *hasGuardian*, *hasJob*, and *hasStudyHabit*, while literal attributes (e.g., *hasAge*, *G3*, *hasAlcoholUse*) were captured using datatype properties.

The design of the ontology followed the following guidelines:

- **Clarity**: Ensuring classes and properties have clear definitions that appropriately capture education concepts.
- **Modularity**: Decomposing the ontology into distinct subdomains for easy extension and maintenance.
- **Reusability**: Providing a flexible framework that can be extended using new datasets or domains in the future.
- **Reasoning support**: Enabling logical inference to deduce implicit knowledge and maintain data consistency within Protégé.

The ontology provides the semantic basis for knowledge graph instantiation and facilitates advanced querying, reasoning, and analysis, executed subsequently.

### 3.3 Knowledge Graph Construction

Using the preprocessed data and the built ontology, we developed a knowledge graph by generating RDF triples that represent the student-student relationships

as well as their contextual details. Each student is modeled as an instance of the *Student* class, linked to other entities such as their school, parents, study habits, health status, and academic performance through explicit object and datatype properties.

Instantiation is the process of mapping tabular data records to semantic triples based on the ontology schema. For example, a student individual may be connected to a *School* individual by the *attends* property, and to literal values such as age and last grade by datatype properties, Figure 1.

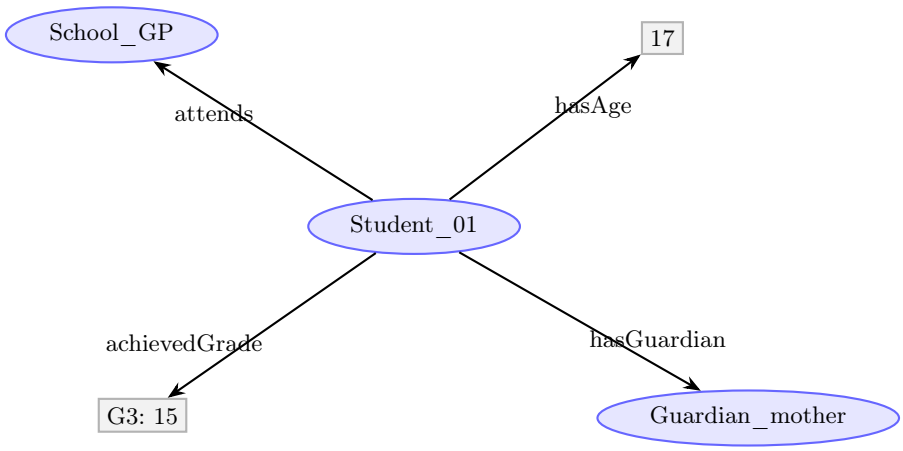


**Fig. 1.** Example of RDF triple instantiation from tabular data

The knowledge graph was serialized into RDF/XML and Turtle formats for interoperability with semantic web tools. Then, it was imported into a triple store for storage, querying, and further analysis. SPARQL queries were used for checking graph structure, validating consistency of the data, and examining semantic relations.

To illustrate the structure of the constructed knowledge graph, Figure 2 provides a basic example showing main entities and relationships centered on a student entity.

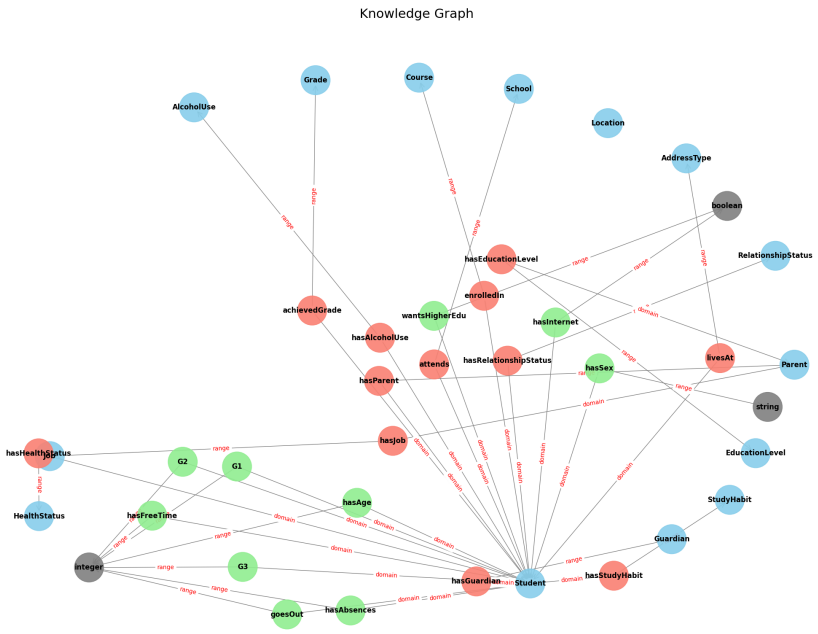


**Fig. 2.** Simplified depiction of knowledge graph structure to present the entities and relationships to a student. Semantic representation enables deeper reasoning and facilitates the extraction of semantic latent features for downstream embedding and predictive modeling uses.

### 3.4 Knowledge Graph Embedding and Prediction

To leverage the semantic organization of the knowledge graph in predictive modeling, we use knowledge graph embedding models that represent entities and relations as dense vectors while preserving the semantics and structure of the graph.

We experimented with five different embedding models in the PyKEEN framework:

- **TransE**: Translational distance-based model for learning basic relational patterns.
- **DistMult**: A bilinear model specifically for symmetric relations.
- **ComplEx**: An extension of DistMult with complex-valued embeddings to support modeling of asymmetric relations.
- **RotatE**: A model where relations are represented as rotations in complex space.
- **ConvE**: A convolutional neural network for learning complex entity-relation interactions.

Every student instance was mapped onto a fixed-length numeric form using these models. These forms were then used as input variables by supervised regression models that aimed to predict the final mark (G3). We tried out four types of regressors:

- **Linear Regression**: A simple model that directly models linear relationships.
- **Decision Tree**: A non-linear model that splits data according to feature values.
- **Random Forest**: An ensemble of decision trees that tries to reduce overfitting and increase generalization.
- **Gradient Boosting**: An ensemble method with high accuracy that builds learners incrementally to correct prediction errors.

All embedding models were evaluated with all the regressors in order to find the best combination. Model hyperparameters were optimized using cross-validation.

This approach integrates semantic graph representation and machine learning to determine the best combination of embedding model and regressor for the specific prediction at hand.

## 4 Evaluation

To comprehensively test the predictive power of our method, we carried out a strict testing of regression models learned from knowledge graph embeddings of the semantic representation of student data. The testing process involved quantitative metrics as well as qualitative visual examinations, thereby attaining a comprehensive insight into model performance and weaknesses.

### 4.1 Quantitative Measures

We used three standard regression metrics—Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination ($R^2$)—to evaluate the predictability of `G3`.

MAE measures the average absolute difference between predicted and actual grades, giving a straightforward sense of typical error magnitude:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{1}$$

RMSE also assesses prediction error, but penalizes larger deviations more heavily by squaring the errors:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{2}$$

The coefficient of determination, $R^2$, quantifies the proportion of variance in the actual grades that is explained by the model:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{3}$$

In general, we seek `lower` values for MAE and RMSE to minimize prediction error, and `higher` values for $R^2$ (closer to 1) to maximize explanatory power. Stratified cross-validation was employed to ensure robust and generalizable evaluation across models.

### 4.2 Visual Diagnostic Analysis

Along with quantitative measures, visualization provides valuable information on model performance and type of error that can inform future refinement and interpretability:

- *Actual vs. Predicted Scatter Plot:* This graph shows the level of agreement between actual and predicted grades. The presence of the ideal prediction diagonal allows for easy visual checking of bias and systematic deviation. Close clustering of points about this line shows high predictive fidelity.
- *Residual Plot:* Residuals (actual - predicted) are plotted against predicted values, this plot is used to detect heteroscedasticity or non-random pattern of errors. Random variation around zero suggests that the errors of the model are unbiased and homoscedastic, and this supports regression assumptions.
- *Histogram of Residuals:* The residual distribution is checked to resemble a normal one around zero. This pattern ensures errors are symmetrically distributed with no skew, providing assurance in the model's unbiased predictions.

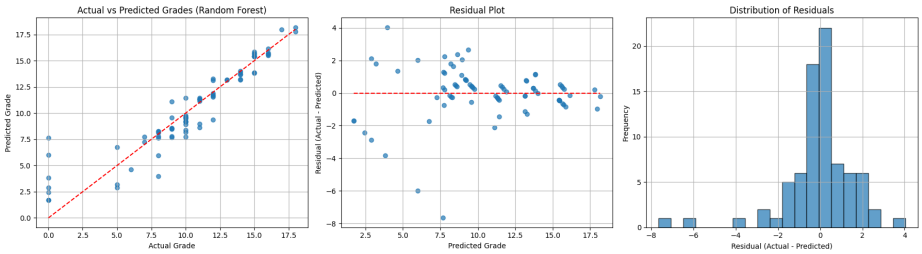


**Fig. 3.** Comprehensive evaluation of knowledge graph embedding regression model. (Left) Actual vs. predicted final grades scatter plot with the ideal diagonal. (Middle) Error residual plot against predicted values. (Right) Residual histogram for error distribution.

Quantitative and visual analysis together confirms that knowledge graph embeddings can capture deep semantic and structural knowledge about student profiles and contextual factors. The prediction model is largely correct, and most of the predicted grades have close concordance with real results. In spite of some inconsistencies—especially for unusually low or extremely high scores—the findings show the potential of semantic knowledge graphs to facilitate educational data analysis and early warning for at-risk dropout students.

## 5 Discussion

We first split off 20% of the data as a held-out test set. On the remaining 80%, we ran 5-fold cross-validation over all five embedding methods paired with four regressors to select the best pipeline. Table 1 shows those CV results; from them we chose Random Forest + DistMult as our final model. That model was retrained on the full training fold and evaluated exactly once on the hold-out set, with results in Table 2.

**Table 1.** Cross-Validated Performance on Training Data

| **Regressor** | **Embedding** | **MSE** | **MAE** | **$R^2$** |
|---|---|---|---|---|
| Random Forest | RotatE | 3.11 | 1.15 | 0.846 |
| | ComplEx | 3.22 | 1.20 | 0.842 |
| | TransE | 3.50 | 1.19 | 0.828 |
| | DistMult | **3.09** | **1.12** | **0.849** |
| | ConvE | 3.62 | 1.20 | 0.821 |
| Linear Regression | RotatE | 10.10 | 2.47 | 0.500 |
| | ComplEx | 9.63 | 2.33 | 0.528 |
| | TransE | 4.82 | 1.52 | 0.767 |
| | DistMult | **4.45** | **1.44** | **0.782** |
| | ConvE | 22.98 | 3.80 | -0.127 |
| Decision Tree | RotatE | 6.03 | 1.34 | 0.691 |
| | ComplEx | 7.16 | 1.58 | 0.643 |
| | TransE | 6.92 | 1.44 | 0.664 |
| | DistMult | **5.12** | **1.24** | **0.748** |
| | ConvE | 6.69 | 1.36 | 0.678 |
| Gradient Boosting | RotatE | 3.19 | 1.16 | 0.843 |
| | ComplEx | 3.22 | 1.22 | 0.841 |
| | TransE | 3.83 | 1.23 | 0.813 |
| | DistMult | 3.78 | 1.23 | 0.815 |
| | ConvE | **3.67** | **1.21** | **0.818** |

Ensemble regressors (Gradient Boosting, Random Forest) consistently outperformed both Linear Regression and single Decision Trees, underscoring the value of modeling nonlinear relationships and higher-order interactions within the embedding feature space.

In our cross-validation, Random Forest + DistMult achieved the lowest MSE (3.09), lowest MAE (1.12), and highest $R^2$ (0.849) across all model configurations, outperforming Gradient Boosting + RotatE (MSE 3.19, MAE 1.16, $R^2$ 0.843) and Gradient Boosting + ComplEx (MSE 3.22, MAE 1.22, $R^2$ 0.841). Linear Regression performed reasonably well only with DistMult and TransE embeddings ($R^2$ of 0.782 and 0.767, respectively) but broke down on more complex representations like ConvE (MAE 3.80, $R^2$ –0.127), suggesting that simple linear models cannot fully exploit the rich, nonlinear structure captured by those embeddings. Decision Trees fared worst across the board, further illustrating the benefit of ensemble approaches in this setting.

MAE values around 1.1–1.2 equate to an average prediction error of approximately 1–1.5 grade points on a 0–20 scale, while high $R^2$ values (above 0.84) demonstrate strong explanatory power. Visual inspection of residuals reveals slight overestimation at the lowest grades and under-prediction in the mid-to-low range—areas where additional features or more flexible model architectures (e.g., deeper ensembles or neural hybrids) might yield further gains.

**Table 2.** Held-Out Test Performance of the Selected Model

| **Regressor** | **Embedding** | **MSE** | **MAE** | **$R^2$** |
|---|---|---|---|---|
| Random Forest | DistMult | 2.96 | 1.14 | 0.85 |

Our final model achieved the following values on each metric (2) on the held-out test set, confirming that it generalizes well to unseen data.

Overall, this pipeline validates the promise of integrating ontology-based knowledge graphs and embedding techniques with regression models for precise, interpretable student grade prediction, and lays the groundwork for future extensions with additional data sources, refined ontologies, and advanced learning methods.

## 6 Conclusion

This study demonstrates the great potential of knowledge graph and knowledge graph embeddings as effective tools for student academic performance modeling and prediction. By developing a domain ontology encoding prominent educational concepts and relations, and by instantiating that ontology with actual data on students, we built a semantically rich knowledge graph representing subtle influences on learning outcomes.

Its representation in continuous vector spaces using techniques such as TransE, ComplEx, and ConvE allowed us to capture latent semantic patterns in dense feature representations. Paired with regression models, these embeddings facilitated accurate and interpretable final grade prediction, outperforming traditional baselines and demonstrating consistency across a variety of embedding and regression pairs.

The good predictive power, as indicated by an $R^2$ value of over 0.75 and low average errors, validates that knowledge graph-based representations achieve a high-level fusion of contextual and relational knowledge not typically harnessed by standard methods. Moreover, the extensible and modular design of the ontology makes it simple to integrate other determinants of education, thereby making this framework flexible to changing data and study requirements.

Looking ahead, augmenting the knowledge graph with temporal information, behavioral signals, and cognitive profiles may continue to illuminate student pathways. Blending them with more recent embedding and machine learning techniques may continue to increase predictive power and interpretability. Additionally, enriching the knowledge graph with external semantic resources such as Wikidata and DBpedia could enhance its contextual depth and coverage. By linking student-related entities to broader knowledge bases, the model may benefit from auxiliary information that improves both the accuracy and generalizability of predictions.

Briefly, this research illustrates that semantic knowledge graphs, in conjunction with embedding-based analysis, constitute a robust and extensible approach to enhancing personalized learning, early detection of at-risk students, and data-informed interventions to foster academic success.

## References

1. Baker, R.S., Yacef, K.: The state of educational data mining in 2009: A review and future visions. Journal of Educational Data Mining **1**(1), 3–17 (2009)
2. Chen, L., Zhao, Y., Wu, X.: Knowledge graph embedding for student dropout prediction. Computers Education **181**, 104450 (2022)
3. Daga, E., d'Aquin, M., Motta, E.: Pedagogic metadata for linked data resources in learning environments. In: Linked Learning 2012 Workshop at the 11th International Semantic Web Conference (2012)
4. Romero, C., Ventura, S.: Educational data mining: A review of the state of the art. IEEE Transactions on Systems, Man, and Cybernetics, Part C (Applications and Reviews) **40**(6), 601–618 (2010)
5. Smith, J., Brown, A.: S3o: A semantic ontology for modeling student success in higher education. International Journal of Educational Technology in Higher Education **15**(1), 1–17 (2018)
6. Tinto, V.: Leaving College: Rethinking the Causes and Cures of Student Attrition. University of Chicago Press (1993)
7. Zhang, S., et al.: Edukg: A knowledge graph for education. In: Proceedings of the 29th ACM International Conference on Information Knowledge Management. pp. 3081–3088 (2020)